

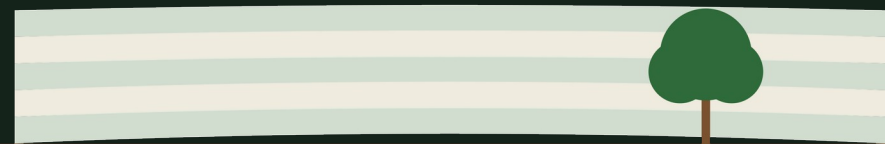
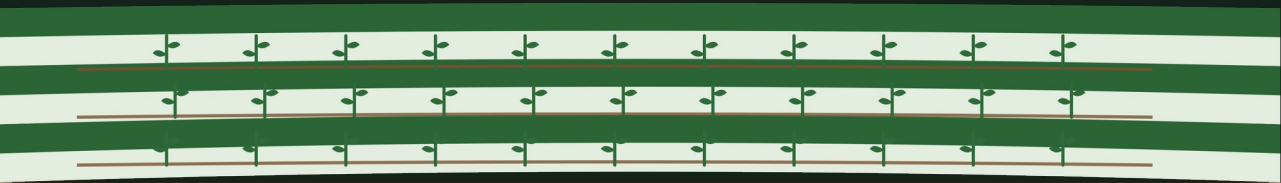
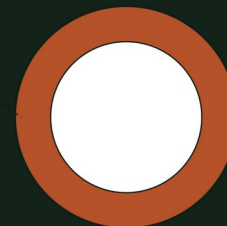
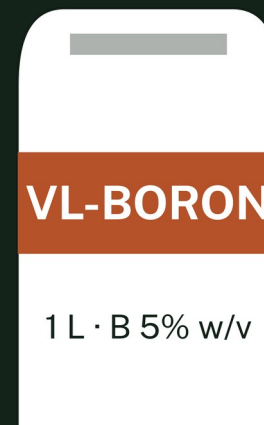


DEALER TRAINING · SEPTEMBER 2026

VL-Boron

Boron 5% w/v, 50 g in a litre, on the leaf or through the water before the flower, at the plan's rate.

For VAN dealers and the field team, written so a dealer can repeat it to a grower. Lahore.



What a dealer says in 30 seconds.

VL-Boron is boron in a bottle: 5% weight in volume, 50 g in every litre, fully soluble, sprayed on the leaf or run through the water. Boron builds the cell wall with calcium and carries sugar to the flower; it decides whether pollen works, and so whether the boll, the head, the panicle or the fruit sets. In most crops it does not move once it is in the plant, so what the flower did not get while it was forming, it never gets. 51.7% of Punjab samples read low in boron. The programmes put 1 litre an acre on at early growth on wheat, sugarcane and canola, and at grand growth on cotton, maize and potato, the weeks before the crop sets. Boron has the narrowest gap of any nutrient between enough and too much, so the plan's rate is the rate. Weigh it, never round it up.

The 4 things it must contain

1

The bottle: 1 L, B 5% w/v, PS 5469-2020. 50 g of boron in every litre. A 20 L pack too.

2

The nutrient: cell wall with calcium, sugar to the flower, pollen and set. 51.7% of Punjab samples read low.

3

The stage: before and at flowering, at the row the plan prints. It does not move inside the plant, so late is too late.

4

The rate: the plan's, exactly. Boron is the 1 nutrient where more is worse.

Say it in the grower's own words. Every figure in these 30 seconds is on the bag, in the published crop programme, or in the named evidence on the slides that follow.

What the bottle says, and what it is registered as.

1L

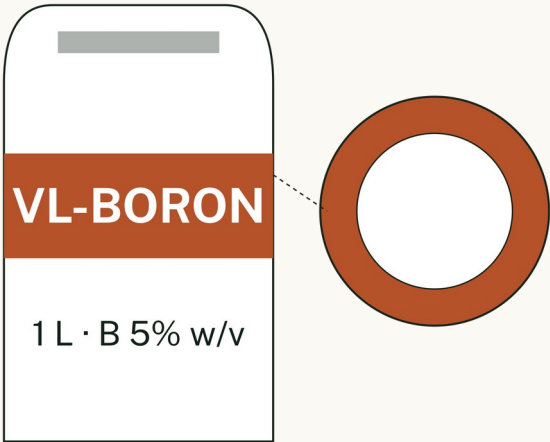
the bottle. 50 g of boron in it, from the declared analysis. 1 litre an acre on most programme rows. A 20 L pack is on the price sheet.

B 5%

boron, w/v: 5 g in every 100 ml. The programme rows print it as "Boron 5%".

Grams

the whole requirement. A crop takes boron in grams an acre, and the gap between short and too much is measured in the same grams.



2 routes

foliar spray and fertigation, as VAN's product data names them. The programme row says which for your crop, and 4 rows print a fraction of a litre.

51.7%

of 770,160 Punjab samples read below 0.5 ppm boron; mean 0.59 ppm.

24

of the 28 published programmes name VL-Boron, 27 rows.

Registered as	Pakistan Standard	Manufacturing licence
VL-Boron	PS 5469-2020	CM/L-3159/2024

No expiry date is printed in this deck. The licence number is what a dealer quotes.

A GROWER'S QUESTION

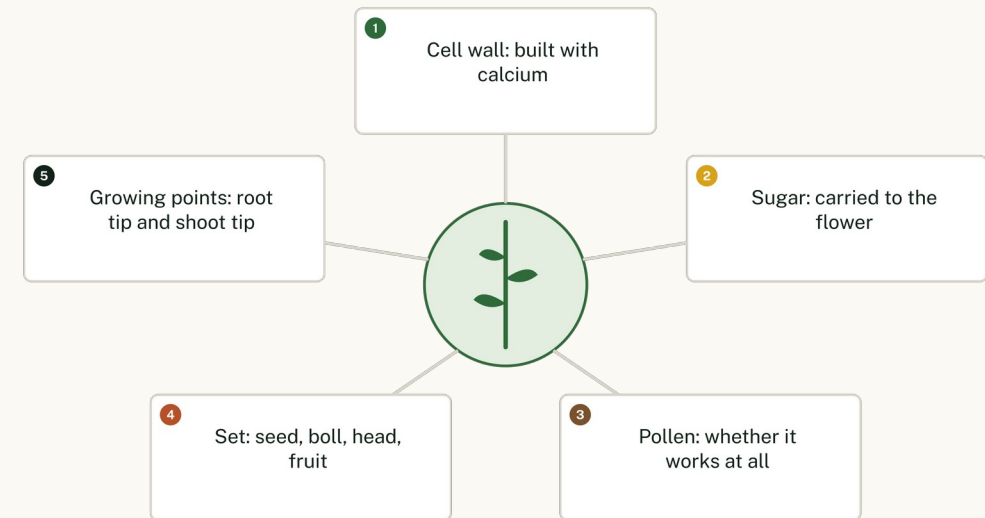
“The cotton grew tall and green and then the squares dropped. The sunflower heads came half empty. The plant looked perfect. What went wrong at flowering?”

The grower did nothing wrong that he could see. The crop built itself well on nitrogen, phosphorus and potash and then failed at the 1 job those 3 do not do: setting. The answer is 3 ideas: what boron does at the flower, why it has to be there before the flower opens, and why the rate is the rate.

The 3 answers on the next 3 slides, 1 idea each.

Boron decides whether the flower sets.

Boron cross-links the cell wall together with calcium, and carries sugar to the flower. Pollen grains and the tube they grow both fail without it. So a crop short of boron can look healthy in leaf and still drop its squares, set a half-empty head, a sterile panicle or a cracked fruit. Growing points die back first, because they are the newest tissue and boron reaches them last. A cotton boll, a sunflower head or a rice panicle that sets badly is often a boron story.



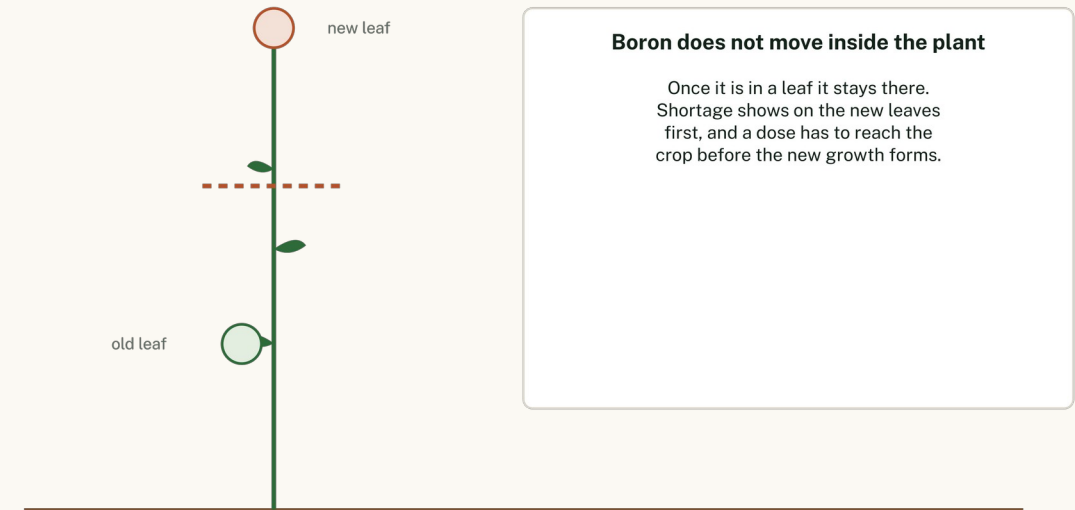
The 5 jobs of boron, from the site's 17-nutrients page. A picture of the idea, not a measurement.

Say it like this: “Nitrogen grew the plant. Boron is what sets the flower. Your crop was short at the 1 moment that decided the yield.”

It does not move inside the plant. Late is too late.

In most crops boron cannot be pulled out of an old leaf and sent to a flower. It moves up with the water the plant drinks and stays where it lands. So the flower has to receive its boron while it is forming, from the soil water or from a spray on the leaf in the weeks before. The programmes put VL-Boron at early growth on wheat, sugarcane, canola and the pulses and at grand growth on cotton, maize, potato and sunflower: before the crop sets, at the row the plan prints. A spray after fruit set is arriving for a job already done.

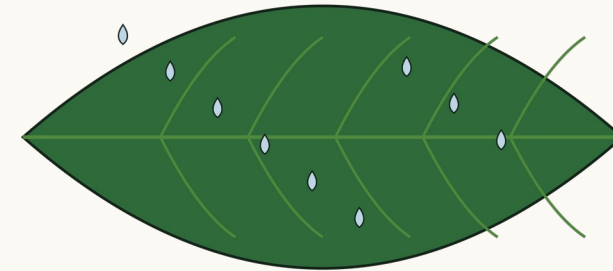
Say it like this: “Put it on before the flower, not after. Once the flower has opened without it, no spray can go back.”



An immobile nutrient: what the old leaf holds, the flower never gets. General plant physiology, stated as such.

Grams, exactly. The plan's rate is the rate.

A crop needs boron in grams an acre, and the gap between short and too much is only a few grams more. Too little costs the flower; too much scorches the leaf edge and costs the plant. That is why VAN sells it as a measured liquid, 50 g in a litre, that can be metered to the gram, and why the programme rows print fractions: $\frac{1}{4}$ litre on chickpea, 200 ml on citrus, 1 litre on cotton. It is also why VAN coats boron onto its potash granules rather than selling it loose in a bag. The plan's figure is not a starting point to round up.



Sprayed onto the leaf. Small doses, taken up through the leaf surface, for a nutrient the crop needs quickly or that the soil locks up. Spray early morning or late afternoon; never on a wilted crop.

The leaf taking up the droplet. The rate comes from the plan; no universal rate is printed.

Say it like this: “50 g in the litre. Measure the plan's amount and stop. This is the 1 nutrient where more is worse.”

What is verified for VL-Boron, and what is not.

No trial of VL-Boron is on VAN’s record. What a dealer may state as fact is the bottle, the licence, the programme rows and the Punjab soil record, each as what it is.

Item	What it says	How a dealer may use it
The bottle	VL-Boron, boron 5% w/v, 50 g per litre, liquid, fully soluble, 1 L. A 20 L pack on the price sheet.	As fact. Grams per litre from the declared analysis.
The licence	Pakistan Standard PS 5469-2020 for liquid nutrient supplements; manufacturing licence CM/L-3159/2024, as the master record holds it.	As fact. The number, never an expiry date.
The programme rows	27 rows on 24 programmes: 1 litre on most; ¼ litre on chickpea, 300 ml on sesame, 200 ml on soybean, a 200 ml dose on citrus; 2 litres on banana; guava at 500 ml per 100 L. Fertigation on 15 rows, spray on 10, either on cotton and maize.	As the published rate. Verbatim. The fractions are the point.
The soil record (Punjab, 770,160 samples)	51.7% of samples below 0.5 ppm boron; mean 0.59 ppm. 69.1% above pH 8.	As fact about the soil. Never as product performance.
Boron on the potash granule	VAN coats boron onto Vital Potash and Fusion Potash in 2 forms, sodium borate fast and calcium borate slow, on its own line since 2014. Named, never numbered.	As the reason VL-Boron and the coated potash are 2 different jobs: the granule covers the season, the bottle covers the stage.
Not on file	A VL-Boron trial; a yield or value-to-cost figure; the 40 to 78% area, 50% Punjab, 92 t against 2,245 t and crop-response percentages printed in earlier VAN material (published research, not VAN’s); tissue ppm thresholds; a drone rate; the 19.5% solid.	Not said until named.

Sources: the bottle; VAN’s published crop programmes on van.com.pk; the farmer price sheet, Aug 2026; the PSQCA master record; the Punjab soil record, 770,160 samples; VAN’s 17-nutrients page and its Vital Potash page; VAN’s own product data.

The bottle in grams, and the fractions the plan prints.

Prices from the VAN farmer price sheet, Aug 2026, delivered farmer price. Grams from the declared analysis, w/v: 5% is 50 g in a litre.

VL-Boron, B 5% w/v	Volume	Boron	Rupees	Which rows
1 bottle	1 L	50 g	1,490	wheat, cotton, maize, potato, sugarcane and most rows
½ bottle	500 ml	25 g	745	guava: 500 ml per 100 L of water
¼ bottle	250 ml	12.5 g	373	chickpea, spray at early growth
200 ml dose	200 ml	10 g	298	citrus (per tree, as printed) and soybean (1/5 pack)
1 pack	20 L	1.0 kg	22,000	1,100 a litre; 20 acres at the 1 L rows

A bottle is 50 g of boron for 1,490 rupees, 1 acre on most rows. The fractions are not a saving; they are the rate. A grower who rounds ¼ bottle up to a whole one on chickpea has put 4 times the boron on a crop that asked for 12.5 g.

VL-Boron 1 L PKR 1,490 and 20 L PKR 22,000: the farmer price sheet, Aug 2026. 50 g = 1,000 ml × 0.05; 12.5 g = 250 × 0.05; 10 g = 200 × 0.05. Fractions as the crop programmes on van.com.pk print them.

What the programme puts on, per acre: the field crops.

Litres per acre from the published programmes; rupees at the VAN farmer price sheet, Aug 2026, 1 L bottles.

Crop	Stage and method, as printed	Litres	Boron	Rupees per acre
Wheat	Early growth, fertigation	1	50 g	1,490
Cotton	Grand growth, spray or fertigation	1	50 g	1,490
Maize	Grand growth, foliar or fertigation	1	50 g	1,490
Sugarcane	Early growth, fertigation	1	50 g	1,490
Sugarcane ratoon	Grand growth, fertigation	1	50 g	1,490
Hybrid rice	Maturity, fertigation	1	50 g	1,490
Canola	Early growth, fertigation	1	50 g	1,490

On wheat the grower spends 1,490 rupees an acre to put 50 g of boron through the water at early growth, the weeks before the head forms. On cotton the same litre goes on at grand growth, before the squares. 1 application each. That is the counter arithmetic; what the set does with it is not a number VAN prints.

VAN's published crop programmes on van.com.pk, rows as printed. Rupees = litres × 1,490, farmer price sheet, Aug 2026.

What the programme puts on, per acre: vegetables, pulses, oilseeds and orchards.

Litres per acre from the published programmes; rupees at the VAN farmer price sheet, Aug 2026, 1 L bottles. 4 rows print less than a bottle.

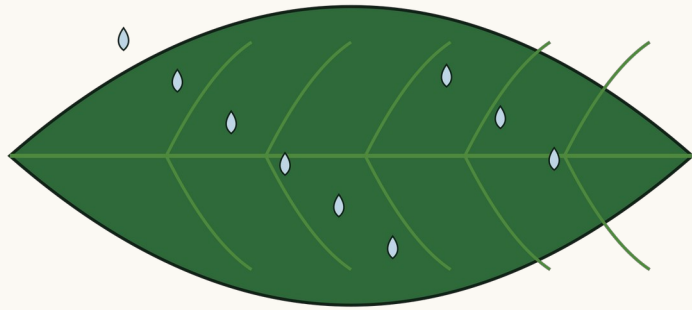
Crop	Stage and method, as printed	Litres	Boron	Rupees per acre
Potato	Grand growth, fertigation	1	50 g	1,490
Tomato	Early growth, spray	1	50 g	1,490
Sunflower	Grand growth, spray	1	50 g	1,490
Chickpea	Early growth, spray	¼	12.5 g	373
Soybean	Early growth, spray	1/5	10 g	298
Garlic	Germination and early growth, fertigation	2	100 g	2,980
Banana, year 1	Germination and grand growth, fertigation, 2 L each	4	200 g	5,960

The pulses take a fraction of a bottle, and the fraction is the rate: chickpea asks for 12.5 g of boron on the acre. Garlic and banana are the rows that take more than 1 application. Every other row here is 1 litre, once.

VAN's published crop programmes on van.com.pk, rows as printed. Rupees = litres × 1,490, farmer price sheet, Aug 2026.

HOW IT GOES ON

On the leaf or through the water, before the flower. Measure it. Water first.



Sprayed onto the leaf. Small doses, taken up through the leaf surface, for a nutrient the crop needs quickly or that the soil locks up. Spray early morning or late afternoon; never on a wilted crop.

Measure the plan's amount, and stop

The row for your crop gives litres or millilitres. Measure it with a jug, never by eye, and never round up. $\frac{1}{4}$ bottle on chickpea means 250 ml. 200 ml on citrus means 200 ml. A guessed boron dose is the commonest error in the field, and boron is the 1 nutrient where the error costs the plant.

Spray: water first, then the bottle

100 to 200 litres of clean water an acre in the tank, then the measured boron, kept agitated. A fine, even fan to cover the leaf without run-off. Morning or evening, never midday. The guava row prints its own water: 500 ml per 100 L. Cotton and maize may go either way; the pulses, sunflower, tomato and citrus are sprayed.

Fertigation: dissolve, then release with the water flowing

Wheat, sugarcane, potato, canola, chili, garlic and banana run it through the water. Into the drum or the drip tank, stirred in, released across the whole irrigation and not in 1 slug. Fully soluble, no residue in the tank or the emitter, as VAN's product data says. Never a drone rate from this deck.

“1,490 for a litre? Borax is a fraction of that at the chemist.”

The dealer says, out loud

Write both on your paper. Then ask how you will measure 12.5 g of boron out of a bag of borax for an acre of chickpea, and get it even across the acre. This bottle is 50 g in a litre, fully soluble, made so the plan's fraction can be poured with a jug and sprayed even. On wheat it is 1 litre, 1,490 rupees, once, at early growth. You are paying for a measured, soluble boron on the 1 nutrient where a wrong measure costs the plant.

The fact behind it

- VL-Boron 1 L is 1,490 rupees; the 20 L pack is 22,000 (farmer price sheet, Aug 2026).
- 50 g of boron in every litre (the declared analysis on the bottle).
- The plan's fractions: ¼ litre on chickpea, 300 ml on sesame, 200 ml on soybean and citrus (the crop programmes on van.com.pk).
- Fully soluble, no undissolved residue in the tank (VAN's product data).

Never say: “It gives you X more maunds” or a value-to-cost ratio. No yield figure exists for the product. No cost per gram of boron against borax.

“My potash already has boron on it. Why buy a bottle too?”

The dealer says, out loud

Because they do 2 jobs. The boron on the Vital Potash and Fusion Potash granule goes in with the potash, in 2 forms, 1 fast and 1 slow, and covers the ground the potash covers. The bottle puts a measured dose on the leaf or in the water in the weeks before the flower, at the row the plan prints. The programme has both on the same crop: the coated potash at its stage, VL-Boron at grand growth on cotton or early growth on wheat. Follow the plan; do not add a bottle the plan does not print, because boron adds up.

The fact behind it

- Vital Potash and Fusion Potash carry boron coated on the granule in 2 forms, sodium borate for fast action and calcium borate for slow release; named, never numbered (the Vital Potash page on van.com.pk).
- VL-Boron and the coated potash sit on the same programmes at different stages (the crop programmes on van.com.pk).
- The gap between enough boron and too much is narrow, which is why VAN coats it on the potash rather than dosing it loose (VAN's 17-nutrients page).
- Boron adds up across products; the plan's rows are the whole dose (textbook agronomy).

Never say: “The granule’s boron is X grams” (never numbered), or “add a bottle anyway to be safe”.

“Will it burn my crop?”

The dealer says, out loud

At the plan's rate, in the plan's water, no. Above it, yes. That is the chemistry of boron, on any label: the gap between enough and too much is a few grams an acre. Burn shows as scorched leaf edges and tips. So measure the row's amount with a jug, put it in 100 to 200 litres of water, and never spray at midday. If your soil test already reads high in boron, do not put it on at all.

The fact behind it

- Too much boron scorches leaf margins and tips; the safe range is narrow (textbook agronomy, as VAN's product guidance states it).
- The gap between enough boron and too much boron is narrow (VAN's 17-nutrients page).
- The plan's rate is measured, never rounded up (VAN's product guidance).
- VAN's general spray guidance: 100 to 200 L of clean water an acre, morning or evening, never midday.

Never say: “It cannot harm the crop” or a rate above the plan's.

“The squares are already dropping. Will a spray now stop it?”

The dealer says, out loud

Boron does not move inside the plant, so the squares that formed without it are gone, and I will not promise a spray brings them back. The cotton row puts 1 litre on at grand growth, before the squares, for exactly this reason. Put it on now for the flowers still to form, at the row's rate, and plan it at grand growth next season. VAN prints no shedding figure for this product.

The fact behind it

- Cotton: 1 litre at grand growth, spray or fertigation (the cotton programme on van.com.pk).
- Boron decides pollination, seed set and fruit set (VAN's 17-nutrients page).
- In most crops boron does not move from old tissue to the flower (general plant physiology).
- No shedding or yield figure exists for VL-Boron.

Never say: “It stops boll drop” or “it sets X% more fruit”.

“Which stage, and how much?”

The dealer says, out loud

What the plan prints, and exactly that. Wheat: 1 litre at early growth, through the water. Cotton and maize: 1 litre at grand growth, spray or water. Sugarcane: 1 litre at early growth; ratoon at grand growth. Potato: 1 litre at grand growth. Canola: 1 litre at early growth. Sunflower: 1 litre at grand growth, sprayed. Chickpea: ¼ litre. Citrus: 200 ml, sprayed, at germination. Garlic: 1 litre twice. Not at maturity, except hybrid rice, which the plan puts there.

The fact behind it

- 24 programmes name VL-Boron, 27 rows (the crop programmes on van.com.pk).
- Early growth on wheat, sugarcane, canola, chili, the pulses, sesame, soybean, tomato, watermelon; grand growth on cotton, maize, potato, sunflower, strawberry, ratoon, turmeric (the same programmes).
- 4 rows print less than a bottle: chickpea ¼, sesame 0.3, soybean 1/5, citrus 200 ml (the same programmes).
- The stage VAN names for the product is mid-life (the VL-Boron page on van.com.pk).

Never say: “1 litre an acre on everything” (4 rows print less), or a rate for a crop with no row.

“Can I put it through the drip on a crop whose row says spray?”

The dealer says, out loud

The bottle takes both routes and is fully soluble, so it will go through the line. But the row's figure is a spray figure for that crop, and boron is the nutrient where the figure matters most. Ask VAN for the fertigation rate for that crop rather than reading the spray bottle into the tank. Where the row prints fertigation, or either, use it as printed.

The fact behind it

- Fertigation on 15 rows, spray on 10, either on cotton and maize (the crop programmes on van.com.pk).
- Foliar spray and fertigation are the 2 routes VAN names for the bottle; fully soluble (VAN's product data).
- A spray row's figure is written for the leaf; a fertigation figure for that crop comes from VAN.

Never say: A fertigation rate the plan does not print.

“Can I put it in my drone?”

The dealer says, out loud

Not from this bottle at this rate. VAN flies a drone grade of VL-Boron, and the site says so, but a drone tank is 20 litres and the concentration is a different specification with its own rate, and boron is the last nutrient to guess a concentration for. Ask VAN for the drone rate.

The fact behind it

- A drone and ultra-low-volume grade of VL-Boron is in production under the same name (the application-systems page on van.com.pk).
- A drone tank carries about 20 litres; everything the crop gets that day is dissolved in it (the same page).
- The knapsack rate is a litre or a fraction in 100 to 200 litres of water; the drone grade has its own rate, from VAN (VAN’s spray guidance).

Never say: “Use the knapsack bottle in the drone.” No drone rate from this deck.

“Is it registered? What is the licence?”

The dealer says, out loud

Pakistan Standard PS 5469-2020 for liquid nutrient supplements, manufacturing licence CM/L-3159/2024, VAN as the registered maker. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. Read the batch number off the bottle and VAN sends you that batch’s certificate, free.

The fact behind it

- VL-Boron: PS 5469-2020, manufacturing licence CM/L-3159/2024 (the regulatory table on van.com.pk, from the PSQCA master record).
- The licence number is printed on the pack; the number is what a dealer quotes, never an expiry date.
- Every batch is released through VAN Lab, PNAC LAB 336, ISO/IEC 17025:2017 (VAN’s quality page).

Never say: An expiry date, or “registered” for a 19.5% boron solid, which this deck does not sell.

VL-Boron in the published programmes, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Early Growth	1 pack	Fertigation	1 L
Maize (Corn)	Grand Growth	1 pack	Foliar or Fertigation	1 L
Cotton	Grand Growth	1 pack	Spray/Fertigation	1 L
Sugarcane, February planting	Early Growth	1 pack	Fertigation	1 L
Ratoon Sugarcane	Grand Growth	1 pack	Fertigation	1 L
Hybrid Rice	Maturity	1 pack	Fertigation	1 L

Rows as published in van.com.pk’s crop programmes, 12 of the 24 programmes that name VL-Boron. 1 litre on every row here except citrus (a 200 ml dose); fertigation on 7, spray on 3, either on cotton and maize. A rate is per acre; the crop page scales it to the field.

VL-Boron in the published programmes, continued.

Crop	Stage	Rate per acre	Method	Pack
Potato	Grand Growth	1 pack	Fertigation	1 L
Tomato	Early Growth	1 pack	Spray	1 L
Chili	Early Growth	1 pack	Fertigation	1 L
Canola	Early Growth	1 pack	Fertigation	1 L
Sunflower	Grand Growth	1 pack	Spray	1 L
Citrus, 5 years and over	Germination	200 ml dose	Spray	1 L

Rows as published in van.com.pk’s crop programmes, 12 of the 24 programmes that name VL-Boron. 1 litre on every row here except citrus (a 200 ml dose); fertigation on 7, spray on 3, either on cotton and maize. A rate is per acre; the crop page scales it to the field.

WHAT VAN WILL NOT CLAIM FOR IT

A dealer who says any of these is speaking for himself, not for VAN.

1 A yield, shedding, sterility or value-to-cost figure. No VL-Boron trial exists. The Pakistani crop-response percentages in earlier VAN material are published research on boron, not on this bottle.

2 That 40 to 78% of field-crop area, or 50% of Punjab soils, is boron-deficient, or that Pakistan uses 92 t against a need of 2,245 t, until a source is on file. The survey's 51.7% below 0.5 ppm is the figure a dealer may say.

3 A universal rate, a rounded-up rate, or a rate for a crop with no programme row. The plan's fraction is the rate.

4 That the boron on the coated potash is X grams, or that the bottle replaces it. Named, never numbered; 2 jobs.

5 A drone rate, a fertigation rate the plan does not print, or tissue ppm thresholds as if VAN had measured them.

6 That the 19.5% solid is registered or for sale. It is a Tier 2 material in registration, off this deck.

Every batch is tested. The bag carries the proof.

1 · The batch number

It is printed on the bag. Read it to the grower and show him where it is. Every bag of every batch carries one.

2 · The certificate of analysis

For any batch, free on request. Send the batch number off the bag to kisan@van.com.pk or the helpline, and the certificate for that batch comes back. No sample to post.

3 · The laboratory

VAN Lab, PNAC LAB 336, accredited to ISO/IEC 17025:2017. Every batch is released through it before it ships. The certificate carries the laboratory's number.

What a certificate for VL-Boron shows

The declared analysis tested for the batch: boron against 5% w/v, with the batch number, the Pakistan Standard PS 5469-2020 and licence CM/L-3159/2024 on the bottle. No certificate for a VL-Boron batch was on file for this deck, so no batch reading is printed here; the grower asks for his own batch's certificate.

The lines a dealer says in Urdu.

The first line is the promise word from the VAN brand master. The other 3 are the deck's key sentences, translated in a first pass and not yet reviewed by VAN. Say them, and send corrections to partner@van.com.pk.

Boron. VAN has no promise word for it; a plain line.

Urdu first pass, unreviewed

بوران پھول کو باندھتا ہے

Put it on before the flower. After the flower it is too late.

Urdu first pass, unreviewed

پھول سے پہلے دیں، پھول کے بعد دیر ہو جاتی ہے

Measure the plan's amount and stop. With boron, more is worse.

Urdu first pass, unreviewed

پلان کی مقدار ناپیں اور بس۔ بوران میں زیادہ نقصان ہے

Water first, then the bottle. Morning or evening.

Urdu first pass, unreviewed

پہلے پانی، پھر بوتل۔ صبح یا شام

Write to the address for what you need.

If you grow

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch, a dealer near you.

If you sell

partner@van.com.pk

Dealers, distributors and the trade. Stock, training, corrections to this deck.

Everything else

info@van.com.pk

Supply, the laboratory, a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore. 1 helpline, 3 addresses. No group broadcasts: a grower who writes gets 1 reply.

VL-Boron: 50 g in the litre, on the leaf or in the water, before the flower, at the plan's rate and no more.

